

Lecture Notes on Elliptic Integrals and Some Applications

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1.1 Introduction to Elliptic Integrals

Elliptic integrals arise naturally in many problems of mathematics and physics where the resulting expressions cannot be written in terms of elementary functions. Historically, they first appeared in the problem of determining the arc length of an ellipse, which leads to integrals that cannot be simplified into combinations of algebraic, exponential, logarithmic, or trigonometric functions.

In general, an elliptic integral is an integral that involves the square root of a polynomial of degree three or four. A typical form is

$$\int R(x, \sqrt{P_4(x)}) dx,$$

where R is a rational function¹ and $P_4(x)$ is a cubic or quartic polynomial which have the form $P_4(x) = a_4 x^4 + a_3 x^3 + a_2 x^2 + a_1 x + a_0$.

Elliptic integrals appear in several important problems in physics. For example, the exact solution of the nonlinear equation of motion of a simple pendulum involves elliptic integrals. Similarly, the arc length of an ellipse leads directly to an elliptic integral of the second kind. Such integrals also arise in problems in classical mechanics, electromagnetism, and orbital dynamics.

Because elliptic integrals cannot be expressed in terms of elementary functions, they form one of the earliest and most important examples of special functions in mathematics. Their study eventually led to the development of elliptic functions, which have deep connections with complex analysis, number theory, and modern mathematical physics.

Among the most commonly encountered elliptic integrals are the complete elliptic integrals of the first and second kind, defined as

$$K(k) = \int_0^{\pi/2} \frac{d\theta}{\sqrt{1 - k^2 \sin^2 \theta}}$$

and

$$K(k) = \int_0^{\pi/2} \sqrt{1 - k^2 \sin^2 \theta} d\theta$$

where k is called the modulus. Before the advent of modern computation, significant effort was devoted to developing analytic techniques and transformations to evaluate such integrals in terms of standard elliptic functions.

With the availability of powerful computers and efficient numerical algorithms, these integrals can now be evaluated directly with high precision. As a result, the practical need for elaborate analytic reduction techniques has decreased. Nevertheless, elliptic integrals remain important in mathematical physics, both for providing exact analytical descriptions of certain systems and for revealing the deeper mathematical structure underlying many problems in classical mechanics, electromagnetism, and geometry.

¹. $R(x, y)$ is a rational function when $R(x, y) = \frac{P(x, y)}{Q(x, y)}$. P, Q are the any functions of x and y . For example consider $R(x, y) = \frac{1}{\sqrt{1-x^4}} = \frac{1}{y}$. Here $y = \sqrt{1-x^4}$ and $P_4(x) = 1 - x^4$.

1.2 Elliptic integral of the first kind

Elliptic integral of the first kind is defined as :

$$F(\varphi \mid \alpha) = \int_0^\varphi (1 - \sin^2 \alpha \sin^2 \theta)^{-\frac{1}{2}} d\theta \quad (1.1)$$

Adopting the standard notation we write, $m = k^2 = \sin^2 \alpha$, we can also write Eq. (1.1) as :

$$F(\varphi, k) = \int_0^\varphi (1 - k^2 \sin^2 \theta)^{-\frac{1}{2}} d\theta \quad (1.2)$$

We can also do this following substitution $t = \sin \theta$ and by putting $x = \sin \varphi$. Using this rewrite Eq. (1.2) as :

$$F(x \mid k) = \int_0^x \left[(1 - t^2)(1 - k^2 t^2) \right]^{-\frac{1}{2}} dt \quad (1.3)$$

In Eqs. (1.2) and (1.3) the value of k is as $0 \leq k \leq 1$ as per definition of k . Now if we take $\varphi = \pi/2 \Rightarrow x = 1$, we have the complete elliptic integral of the first kind :

$$K(k) = \int_0^{\pi/2} (1 - k^2 \sin^2 \theta)^{-\frac{1}{2}} d\theta \quad (1.4a)$$

$$= \int_0^1 \left[(1 - t^2)(1 - k^2 t^2) \right]^{-\frac{1}{2}} dt \quad (1.4b)$$

Now in order to find the value, we do a Binomial expansion of the integrand as follows:

$$(1 - k^2 \sin^2 \theta)^{-\frac{1}{2}} = \sum_{r=0}^{\infty} K_r (k^2 \sin^2 \theta)^r$$

Where $K_0 = 1$ and for other values of r we have :

$$K_r = \frac{(2r-1)!!}{(2r)!!} \quad (1.5)$$

Using this we can expand the integrand as a power series and using this series Eq. (1.4a) takes the form :

$$K(k) = \int_0^{\pi/2} (1 - k^2 \sin^2 \theta)^{-\frac{1}{2}} d\theta$$

2. From the binomial expansion we know that :

$$(1+x)^n = 1 + \sum_{r=1}^{\infty} \binom{n}{r} x^r$$

Where

$$\binom{n}{r} = \frac{n!}{r!(n-r)!} = \frac{n(n-1) \cdots (n-r+1)}{r!}$$

Using this we try to find the expansion of $(1-y)^{-1/2}$ as follows ($n = -\frac{1}{2}$) :

$$(1-y)^{-1/2} = 1 + \sum_{r=1}^{\infty} c_r (-1)^r y^r = 1 + \sum_{r=1}^{\infty} \frac{(2r-1)!!}{(2r)!!} y^r$$

Where :

$$\begin{aligned} c_r &= \frac{\left(-\frac{1}{2}\right) \cdot \left(-\frac{1}{2}-1\right) \cdots \left(-\frac{1}{2}-r+1\right)}{r!} \\ &= \frac{(-1) \cdot (-3) \cdots (1-2r)}{2^r r!} \leftarrow \text{taking } \left(-\frac{1}{2}\right) \text{ common from each terms (total } r \text{ terms.)} \\ &= (-1)^r \frac{(2r-1) \cdot (2r-3) \cdots 3 \cdot 1}{2^r r!} = (-1)^r \frac{(2r-1)!!}{(2r)!!} \end{aligned}$$

Also $2^r r! = 2^r [r(r-1)(r-2) \cdots 2 \cdot 1] = [2r] \cdot [2(r-1)] \cdot [2(r-2)] \cdots [2 \cdot 2] \cdot [2 \cdot 1] = (2r) \cdot (2r-2) \cdots (4) \cdot 2 = (2r)!!$

$$\begin{aligned}
&= \int_0^{\pi/2} d\theta \left[1 + \sum_{r=1}^{\infty} \frac{(2r-1)!!}{(2r)!!} (k^2 \sin^2 \theta)^r \right] \\
&= \int_0^{\pi/2} d\theta + \left[\sum_{r=1}^{\infty} \frac{(2r-1)!!}{(2r)!!} k^{2r} \int_0^{\pi/2} \sin^{2r} \theta d\theta \right] \\
&= \frac{\pi}{2} + \frac{\pi}{2} \left[\sum_{r=1}^{\infty} \left(\frac{(2r-1)!!}{(2r)!!} \right)^2 k^{2r} \right] \\
&= \frac{\pi}{2} \left[1 + \sum_{r=1}^{\infty} \left(\frac{(2r-1)!!}{(2r)!!} \right)^2 k^{2r} \right]
\end{aligned}$$

Now expanding the last equation we got :

$$K(k) = \frac{\pi}{2} \left[1 + \left(\frac{1}{2} \right)^2 k^2 + \left(\frac{1 \cdot 3}{2 \cdot 4} \right)^2 k^4 + \left(\frac{1 \cdot 3 \cdot 5}{2 \cdot 4 \cdot 6} \right)^2 k^6 + \dots \right] \quad (1.6)$$

Now from Eq. (1.6) we get for $k = 0$, we have $K(k = 0) = \frac{\pi}{2}$. That is expected as if we put $k = 0$ in Eq. (1.4a) we get :

$$K(k = 0) = \int_0^{\pi/2} d\theta = \frac{\pi}{2}$$

One thing is that the series we got in Eq. (1.4a) is divergent for $k = 1$. In fact it is “logarithmically divergent” which can be shown as follows :

$$\begin{aligned}
K(k = 1) &= \int_0^{\pi/2} (1 - \sin^2 \theta)^{-\frac{1}{2}} d\theta \\
&= \int_0^{\pi/2} (\cos^2 \theta)^{-\frac{1}{2}} d\theta \\
&= \int_0^{\pi/2} \frac{d\theta}{\cos \theta} \\
&= \int_0^{\pi/2} \sec \theta d\theta \\
&= \ln(\sec \theta + \tan \theta) \Big|_0^{\pi/2} = \ln \infty = \infty
\end{aligned}$$

To summarize we have :

$$\lim_{k \rightarrow 0} K(k) = \frac{\pi}{2} \quad (1.7a)$$

$$\lim_{k \rightarrow 1} K(k) = \infty \quad (1.7b)$$

1.3 Elliptic integral of the second kind

After studying Elliptic integral of the first kind, we now shift our attention to Elliptic integral of the second kind, which is defined as :

$$E(\varphi \mid \alpha) = \int_0^{\varphi} (1 - \sin^2 \alpha \sin^2 \theta)^{\frac{1}{2}} d\theta \quad (1.8)$$

If we do same kind of substitution as we did before we end up getting :

$$E(x \mid k) = \int_0^x \left[\frac{1 - k^2 t^2}{1 - t^2} \right]^{\frac{1}{2}} dt \quad (1.9)$$

³. We have used the value of integral which is done in Sec. 1.7.

Where the same symbols hold. Similarly we have the complete elliptic integral of the second kind :

$$E(k) = \int_0^{\pi/2} (1 - k^2 \sin^2 \theta)^{\frac{1}{2}} d\theta \quad (1.10a)$$

$$= \int_0^1 \left[\frac{1 - k^2 t^2}{1 - t^2} \right]^{\frac{1}{2}} dt \quad (1.10b)$$

Binomial expansion of the integrand gives :

$$(1 - k^2 \sin^2 \theta)^{\frac{1}{2}} = 1 - \sum_{r=1}^{\infty} \frac{(2r-1)!!}{(2r)!!} \frac{(k^2 \sin^2 \theta)^r}{2r-1}$$

Finally we have :

$$\begin{aligned} E(k) &= \int_0^{\pi/2} (1 - k^2 \sin^2 \theta)^{\frac{1}{2}} d\theta \\ &= \int_0^{\pi/2} d\theta \left[1 - \sum_{r=1}^{\infty} \frac{(2r-1)!!}{(2r)!!} \frac{(k^2 \sin^2 \theta)^r}{2r-1} \right] \\ &= \int_0^{\pi/2} d\theta - \left[\sum_{r=1}^{\infty} \frac{(2r-1)!!}{(2r)!!} \frac{k^{2r}}{2r-1} \int_0^{\pi/2} \sin^{2r} \theta d\theta \right] \\ &= \frac{\pi}{2} - \frac{\pi}{2} \left[\sum_{r=1}^{\infty} \left(\frac{(2r-1)!!}{(2r)!!} \right)^2 \frac{k^{2r}}{2r-1} \right] \\ &= \frac{\pi}{2} \left[1 - \sum_{r=1}^{\infty} \left(\frac{(2r-1)!!}{(2r)!!} \right)^2 \frac{k^{2r}}{2r-1} \right] \end{aligned}$$

Expansion gives :

$$E(k) = \frac{\pi}{2} \left[1 - \left(\frac{1}{2} \right)^2 \frac{k^2}{1} - \left(\frac{1 \cdot 3}{2 \cdot 4} \right)^2 \frac{k^4}{3} - \left(\frac{1 \cdot 3 \cdot 5}{2 \cdot 4 \cdot 6} \right)^2 \frac{k^6}{5} - \dots \right] \quad (1.11)$$

Again we have :

$$E(k=0) = \int_0^{\pi/2} d\theta = \frac{\pi}{2}$$

and for $k=1$ we have :

$$\begin{aligned} E(k=1) &= \int_0^{\pi/2} (1 - \sin^2 \theta)^{\frac{1}{2}} d\theta \\ &= \int_0^{\pi/2} (\cos^2 \theta)^{\frac{1}{2}} d\theta \\ &= \int_0^{\pi/2} d\theta \cos \theta \end{aligned}$$

4. Similarly the expansion of $(1-y)^{1/2}$ is given by $(n = \frac{1}{2})$:

$$(1-y)^{1/2} = 1 + \sum_{r=1}^{\infty} d_r (-1)^r y^r = 1 - \sum_{r=1}^{\infty} \frac{(2r-1)!!}{(2r)!!} \frac{y^2}{(2r-1)}$$

Where :

$$\begin{aligned} d_r &= \frac{\left(\frac{1}{2}\right) \cdot \left(\frac{1}{2}-1\right) \cdots \left(\frac{1}{2}-r+1\right)}{r!} \\ &= \frac{(1) \cdot (-1) \cdot (-3) \cdots (3-2r)}{2^r r!} \leftarrow \text{taking } \left(-\frac{1}{2}\right) \text{ common from each terms (total } r \text{ terms.)} \\ &= -(-1)^r \frac{(2r-3) \cdot (2r-5) \cdots 3 \cdot 1}{2^r r!} = -(-1)^r \frac{(2r-1)!!}{(2r)!!} \cdot \frac{1}{(2r-1)} \end{aligned}$$

5. See Sec. 1.7.

$$= \sin \theta \Big|_0^{\pi/2} = 1$$

To summarize we have :

$$\lim_{k \rightarrow 0} E(k) = \frac{\pi}{2} \quad (1.12a)$$

$$\lim_{k \rightarrow 1} E(k) = 1 \quad (1.12b)$$

Even for $k = 1$, $E(k)$ convergent. The values of $K(k)$ and $E(k)$ are numerically computed and we have plotted them in Fig (1.1).

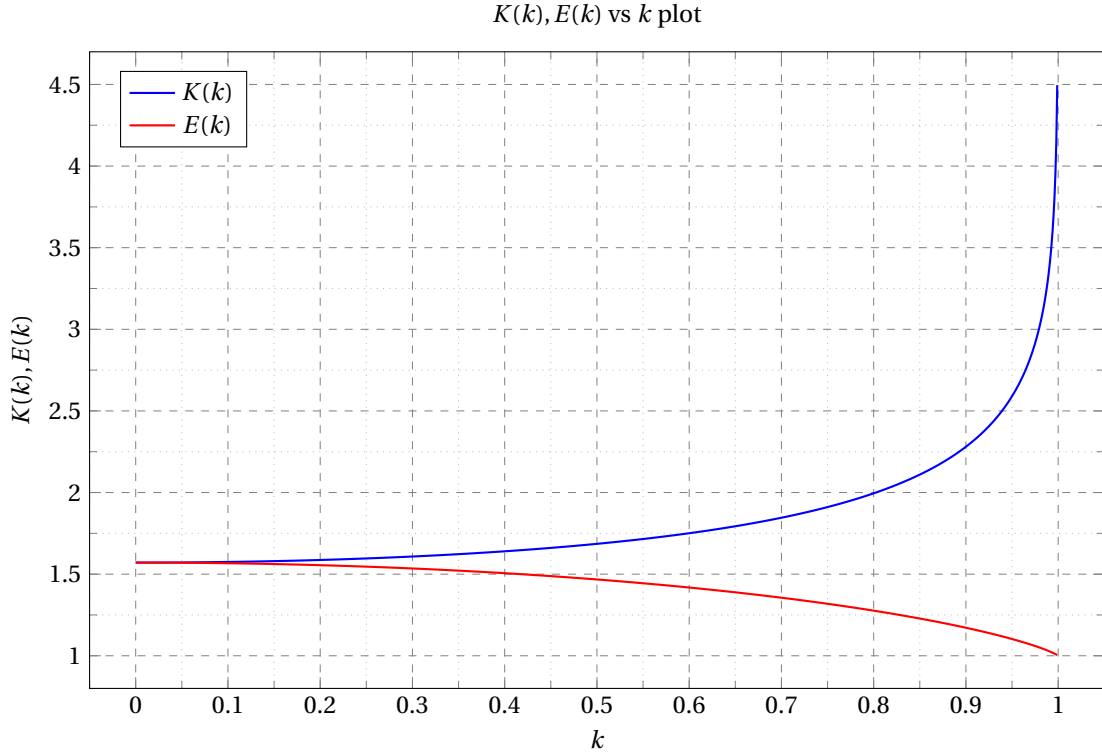


Figure 1.1: Plot of $K(k)$ and $E(k)$ as a function of k in the range $0 \leq k \leq 1$. For $K(k)$ we avoided getting near $k \rightarrow 1$. We have computed $K(k)$ and $E(k)$ using the discussions made in Sec. 1.6.2.

1.4 Writing $K(k)$ and $E(k)$ as power series in k

Following Eq. (1.6) we write :

$$K(k) = \frac{\pi}{2} \left[1 + \sum_{r=1}^{\infty} \left(\frac{(2r-1)!!}{(2r)!!} \right)^2 k^{2r} \right] = \sum_{r=1}^{\infty} a_r k^{2r} \quad (1.13)$$

Where we have :

$$a_0 = \frac{\pi}{2}, \quad a_r = \frac{\pi}{2} \left(\frac{(2r-1)!!}{(2r)!!} \right)^2, r \geq 1 \quad (1.14)$$

Using this we can easily get :

$$\begin{aligned} \frac{a_{r+1}}{a_r} &= \left[\frac{(2r+1)!!}{(2r+2)!!} \times \frac{(2r)!!}{(2r-1)!!} \right]^2 \\ &= \left[\frac{(2r+1) \cdot \cancel{(2r-1)!!}}{(2r+2) \cdot \cancel{(2r)!!}} \times \frac{\cancel{(2r)!!}}{\cancel{(2r-1)!!}} \right]^2 \\ &= \left[\frac{2r+1}{2r+2} \right]^2 \end{aligned}$$

From which we get the recursion relation for a_r , given by :

$$a_{r+1} = \left[\frac{2r+1}{2r+2} \right]^2 a_r \quad (1.15)$$

The last equation holds for all r values as you can easily check by putting $r = 0$, which gives $a_1 = \frac{a_0}{4}$. We have used this recursion relation for the evaluation of $K(k)$ instead of computing double factorial for each coefficients.

In a very similar sense, following Eq. (1.11) we write :

$$E(k) = \frac{\pi}{2} \left[1 - \sum_{r=1}^{\infty} \left(\frac{(2r-1)!!}{(2r)!!} \right)^2 \frac{k^{2r}}{2r-1} \right] = \sum_{r=1}^{\infty} b_r k^{2r} \quad (1.16)$$

Where we have :

$$b_0 = \frac{\pi}{2}, \quad b_r = -\frac{\pi}{2} \left(\frac{(2r-1)!!}{(2r)!!} \right)^2 \frac{1}{2r-1}, r \geq 1 \quad (1.17)$$

Using this we can easily get :

$$\begin{aligned} \frac{b_{r+1}}{b_r} &= \left[\frac{(2r+1)!!}{(2r+2)!!} \times \frac{(2r)!!}{(2r-1)!!} \right]^2 \times \frac{2r-1}{2r+1} \\ &= \left[\frac{(2r+1) \cdot \cancel{(2r-1)!!}}{(2r+2) \cdot \cancel{(2r)!!}} \times \frac{\cancel{(2r)!!}}{\cancel{(2r-1)!!}} \right]^2 \times \frac{2r-1}{2r+1} \\ &= \left[\frac{2r+1}{2r+2} \right]^2 \times \frac{2r-1}{2r+1} \\ &= \left[\frac{(2r+1)(2r-1)}{(2r+2)^2} \right] \end{aligned}$$

Recursion relation for b_r , given by :

$$b_{r+1} = \left[\frac{(2r+1)(2r-1)}{(2r+2)^2} \right] b_r \quad (1.18)$$

The last equation holds for all r values as, by putting $r = 0$, which gives $b_1 = -\frac{b_0}{4}$. We used this for the evaluation of $E(k)$ instead of computing double factorial for each coefficients.

Following Eqs. (1.6) and (1.11), we quote the expansion of $K(k)$ and $E(k)$ as follows :

$$K(k) = \frac{\pi}{2} \left[1 + \frac{1}{4} k^2 + \frac{9}{64} k^4 + \frac{25}{256} k^6 + \dots \right] \quad (1.19a)$$

$$E(k) = \frac{\pi}{2} \left[1 - \frac{1}{4} k^2 - \frac{3}{64} k^4 - \frac{5}{256} k^6 - \dots \right] \quad (1.19b)$$

1.5 Few Applications

1.5.1 Finding the arc length of an Ellipse

Consider an ellipse in “ xy ” plane, which is described by :

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \quad (1.20)$$

We wish to find the “length of the arc” within the “first quadrant”. Now two cases may arise, namely (i) $a > b$ and (ii) $a < b$. We will deal with them separately.⁶

⁶ You might be thinking why is that, for now hold on, you will see in a moment.

Case I : $a > b$

Consider the first case where $a > b$. In Fig (1.2), we have drawn an ellipse for this case. We take the parametric form of x and y as follows :

$$x = a \sin \theta, \quad y = b \cos \theta \quad (1.21)$$

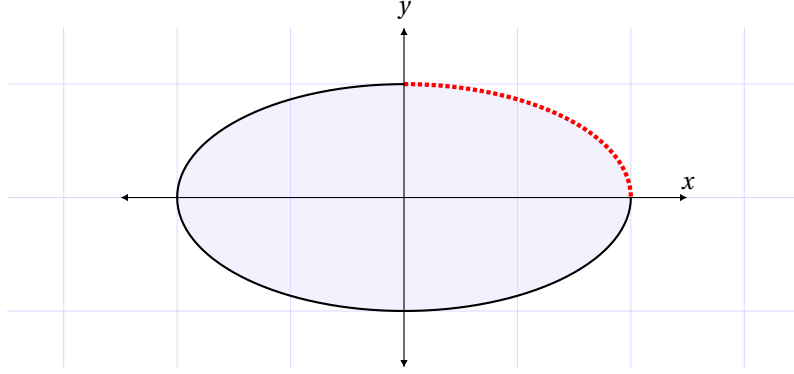


Figure 1.2: Ellipse in xy plane for which $a > b$. Here for convenience we have taken $a = 2\text{cm}$ and $b = 1\text{cm}$. The red dotted line shows the arc within first quadrant.

Now the line element is given by :

$$\begin{aligned} ds &= \sqrt{dx^2 + dy^2} \\ &= \sqrt{a^2 \cos^2 \theta + b^2 \sin^2 \theta} d\theta \\ &= a \sqrt{1 - \sin^2 \theta + \frac{b^2}{a^2} \sin^2 \theta} d\theta \\ &= a \sqrt{1 - \left(1 - \frac{b^2}{a^2}\right) \sin^2 \theta} d\theta \\ &= a \sqrt{1 - k^2 \sin^2 \theta} d\theta \end{aligned}$$

Where we wrote $k^2 = 1 - \frac{b^2}{a^2}$. Now for $a > b$ we have the value of $k < 1$. Now if we had $a < b$, then surely k^2 becomes negative and all the things will not hold any more. We will deal with this very soon. The arc length is given by :

$$S = \int_{\text{F.Q.}} ds = \int_0^{\pi/2} a \sqrt{1 - k^2 \sin^2 \theta} d\theta = a E(k) \quad (1.22)$$

Case II : $b > a$

Fig (1.3) shows an typical ellipse for which $b > a$. Here we take the parametric form :

$$x = a \cos \theta, \quad y = b \sin \theta \quad (1.23)$$

Similarly line element is given by :

$$\begin{aligned} ds &= \sqrt{dx^2 + dy^2} \\ &= \sqrt{a^2 \sin^2 \theta + b^2 \cos^2 \theta} d\theta \end{aligned}$$

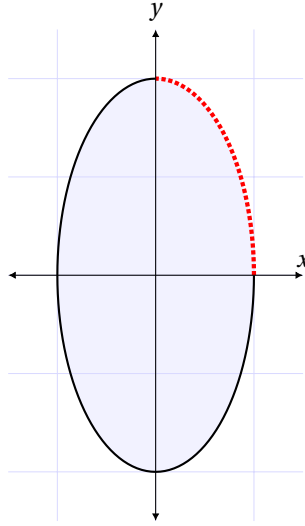


Figure 1.3: Ellipse in xy plane for which $b > a$. Here for convenience we have taken $a = 1\text{cm}$ and $b = 2\text{cm}$. The red dotted line shows the arc within first quadrant.

$$\begin{aligned}
 &= b \sqrt{1 - \sin^2 \theta + \frac{a^2}{b^2} \sin^2 \theta} d\theta \\
 &= b \sqrt{1 - \left(1 - \frac{a^2}{b^2}\right) \sin^2 \theta} d\theta \\
 &= b \sqrt{1 - \tilde{k}^2 \sin^2 \theta} d\theta
 \end{aligned}$$

Here $\tilde{k}^2 = 1 - \frac{a^2}{b^2}$. And as $b < a \Rightarrow \tilde{k} < 1$. The same arguments will hold that was discussed for **Case -I**.

$$S = \int_{\text{F.Q.}} ds = \int_0^{\pi/2} b \sqrt{1 - \tilde{k}^2 \sin^2 \theta} d\theta = b E(\tilde{k}) \quad (1.24)$$

1.5.2 Period of a Simple Pendulum

This problem is not very simple as it sounds. Consider a simple pendulum of mass m and length l suspended from point of suspension O , performing oscillations in the vertical plane. At the horizontal plane passing through the point of suspension, we take potential energy to be zero. i.e. $V = 0$. We can describe the motion of the pendulum in plane polar coordinates, where for the position of the bob can be described by angle θ measured from the vertical plane. This is shown in Fig (1.4).

We set up the Lagrangian as follows :

$$\mathcal{L} = \frac{1}{2} ml^2 \dot{\theta}^2 + mgl \cos \theta \quad (1.25)$$

Using “Euler Lagrange” equation :

$$\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{\theta}} \right) - \frac{\partial \mathcal{L}}{\partial \theta} = 0$$

We arrive at :

$$\ddot{\theta} + \omega^2 \sin \theta = 0 \quad (1.26)$$

Where $\omega = \sqrt{\frac{g}{l}}$. Now Eq. (1.26) is a non linear equation and is hard to solve analytically by all means. What we do often is that, for small amplitude oscillations we approximate $\sin \theta \approx \theta$, which gives Eq. (1.26) as :

$$\ddot{\theta} + \omega^2 \theta = 0 \quad (1.27)$$

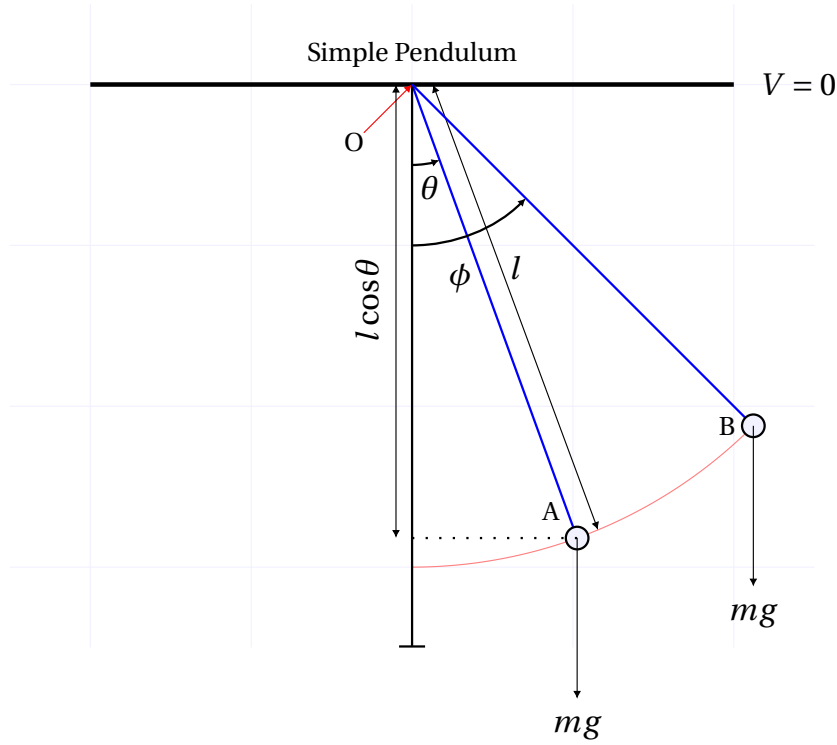


Figure 1.4: Simple Pendulum of mass m and length l suspended from pt “O”. Pt “B” is the maximum angle (ϕ) it can achieve. Pt “A” is a generic point on the path. Thin red line shows the arc for fixed l .

The solutions of Eq. (1.27) is very well known, given by :

$$\theta(t) = A \cos \omega t + B \sin \omega t$$

Which gives time period, $T = \frac{2\pi}{\omega} = 2\pi \sqrt{\frac{l}{g}}$. Which is not practical for large angles. We want to find the time period of the simple pendulum in general. We go back to Eq. (1.26) and by multiplying by $\dot{\theta}$ we get :

$$\begin{aligned} \dot{\theta} \ddot{\theta} + \omega^2 \dot{\theta} \sin \theta &= 0 \\ \Rightarrow \frac{d}{dt} \left(\frac{\dot{\theta}^2}{2} \right) - \omega^2 \frac{d}{dt} (\cos \theta) &= 0 \\ \Rightarrow \frac{d}{dt} \left[\frac{\dot{\theta}^2}{2} - \omega^2 \cos \theta \right] &= 0 \\ \Rightarrow \frac{\dot{\theta}^2}{2} - \omega^2 \cos \theta &= \text{Const} \end{aligned}$$

Now this constant can be found out by using boundary conditions. Say at $t = 0$, the bob starts from $\theta = 0$ and after time $t = t_{1/4}$ ⁷ it reaches maximum displacement (position B), say $\theta = \phi$. At this position the whole energy is purely potential and this gives the value of $\text{Const} = -\omega^2 \cos \phi$. So finally we have :

$$\dot{\theta}^2 = \left(\frac{d\theta}{dt} \right)^2 = 2\omega^2 (\cos \theta - \cos \phi)$$

Or equivalently we have :

$$\frac{d\theta}{dt} = \pm \omega \sqrt{2(\cos \theta - \cos \phi)} \quad (1.28)$$

Here “+” sign signifies where $\dot{\theta} > 0$ and for the “-” sign, $\dot{\theta} < 0$. We will choose the sign accordingly. Now one thing worth

⁷ We have attached a subscript “ $\frac{1}{4}$ ” to signify this time is nothing but the quarter of Time period T .

noting is that we could have obtained Eq. (1.28) straight away by applying Conservation of Energy at point “A” and “B” respectively which we will see very briefly. Conservation of Energy gives :

$$\begin{aligned}
 \text{Energy at Point A} &= \text{Energy at Point B} \\
 \frac{1}{2} m l^2 \dot{\theta}^2 - m g l \cos \theta &= -m g l \cos \phi \\
 \Rightarrow \frac{1}{2} m l^2 \dot{\theta}^2 &= m g l (\cos \theta - \cos \phi) \\
 \Rightarrow \dot{\theta}^2 &= \frac{2g}{l} (\cos \theta - \cos \phi) = 2\omega^2 (\cos \theta - \cos \phi)
 \end{aligned}$$

From Eq. (1.28) we get :

$$dt = \pm \frac{d\theta}{\omega \sqrt{2(\cos \theta - \cos \phi)}}$$

Integrating which from $t = 0$ to $t = t_{1/4}$ or equivalently $\theta = 0$ to $\theta = \phi$, we get :

$$\begin{aligned}
 \int_0^{t_{1/4}} dt &= \int_0^\phi \frac{d\theta}{\omega \sqrt{2(\cos \theta - \cos \phi)}} \quad 8 \\
 \Rightarrow t_{1/4} &= \int_0^\phi \frac{d\theta}{\omega \sqrt{2(\cos \theta - \cos \phi)}}
 \end{aligned} \tag{1.29}$$

As $\theta \leq \phi$ for the range of our interest we do a substitution as follows :

$$\sin\left(\frac{\theta}{2}\right) = \sin\left(\frac{\phi}{2}\right) \sin \alpha \tag{1.30}$$

Due to this the limits are, $\theta = 0 \Rightarrow \alpha = 0$ and $\theta = \phi \Rightarrow \alpha = \pi/2$. Differentiating Eq. (1.30) gives :

$$\begin{aligned}
 \cos\left(\frac{\theta}{2}\right) \frac{d\theta}{2} &= \sin\left(\frac{\phi}{2}\right) \cos \alpha d\alpha \\
 \Rightarrow d\theta &= \frac{2 \sin\left(\frac{\phi}{2}\right) \cos \alpha d\alpha}{\cos\left(\frac{\theta}{2}\right)} \\
 &= \frac{2 \sin\left(\frac{\phi}{2}\right) \cos \alpha d\alpha}{\sqrt{1 - \sin^2\left(\frac{\theta}{2}\right)}} = \frac{2 \sin\left(\frac{\phi}{2}\right) \cos \alpha d\alpha}{\sqrt{1 - \sin^2\left(\frac{\phi}{2}\right) \sin^2 \alpha}}
 \end{aligned}$$

While the denominator gives :

$$\begin{aligned}
 \frac{1}{\omega \sqrt{2(\cos \theta - \cos \phi)}} &= \frac{1}{\omega \sqrt{2\left[1 - 2 \sin^2\left(\frac{\theta}{2}\right) - \cos \phi\right]}} \\
 &= \frac{1}{\omega \sqrt{2\left[1 - 2 \sin^2\left(\frac{\phi}{2}\right) \sin^2 \alpha - \cos \phi\right]}} \\
 &= \frac{1}{\omega \sqrt{2\left[1 - 2 \sin^2\left(\frac{\phi}{2}\right)(1 - \cos^2 \alpha) - \cos \phi\right]}} \\
 &= \frac{1}{\omega \sqrt{2\left[1 - 2 \sin^2\left(\frac{\phi}{2}\right) + 2 \sin^2\left(\frac{\phi}{2}\right) \cos^2 \alpha - \cos \phi\right]}}
 \end{aligned}$$

8. We have taken “+” sign here as in this range $\dot{\theta} > 0$.

$$\begin{aligned}
&= \frac{1}{\omega \sqrt{2 \left[\cancel{\cos \phi} + 2 \sin^2 \left(\frac{\phi}{2} \right) \cos^2 \alpha - \cancel{\cos \phi} \right]}} \\
&= \frac{1}{2\omega \sin \left(\frac{\phi}{2} \right) \cos \alpha}
\end{aligned}$$

Putting it all together in Eq. (1.29) we have :

$$\begin{aligned}
\Rightarrow t_{1/4} &= \int_0^\phi \frac{d\theta}{\omega \sqrt{2(\cos \theta - \cos \phi)}} \\
&= \int_0^{\pi/2} \frac{1}{\cancel{2\omega \sin \left(\frac{\phi}{2} \right) \cos \alpha}} \cdot \frac{\cancel{2 \sin \left(\frac{\phi}{2} \right) \cos \alpha} d\alpha}{\sqrt{1 - \sin^2 \left(\frac{\phi}{2} \right) \sin^2 \alpha}} \\
&= \frac{1}{\omega} \int_0^{\pi/2} \left[1 - \sin^2 \left(\frac{\phi}{2} \right) \sin^2 \alpha \right]^{-1/2} d\alpha \\
&= \frac{1}{\omega} K \left(\sin^2 \frac{\phi}{2} \right)
\end{aligned}$$

So using this we have the time period as :

$$\begin{aligned}
T &= 4t_{1/4} \\
&= \frac{4}{\omega} K \left(\sin^2 \frac{\phi}{2} \right) \\
&= \frac{4}{\omega} \cdot \frac{\pi}{2} \left[1 + \frac{1}{4} \sin^2 \frac{\phi}{2} + \frac{9}{64} \sin^4 \frac{\phi}{2} + \frac{25}{256} \sin^6 \frac{\phi}{2} + \dots \right]
\end{aligned} \tag{1.31}$$

$$\boxed{T = 2\pi \sqrt{\frac{l}{g}} \left[1 + \frac{1}{4} \sin^2 \frac{\phi}{2} + \frac{9}{64} \sin^4 \frac{\phi}{2} + \frac{25}{256} \sin^6 \frac{\phi}{2} + \dots \right]} \tag{1.32}$$

From Eq. (1.32) it is clear that the time period T is Written as a power series of $\sin^2 \frac{\phi}{2}$. Now if we take the “small angle approximation”, i.e. $\sin^2 \frac{\phi}{2} \approx \left(\frac{\phi}{2} \right)^2$, then Eq. (1.32) becomes :

$$\begin{aligned}
T_{\text{approx}} &= 2\pi \sqrt{\frac{l}{g}} \left[1 + \frac{1}{4} \left(\frac{\phi}{2} \right)^2 + \frac{9}{64} \left(\frac{\phi}{2} \right)^4 + \frac{25}{256} \left(\frac{\phi}{2} \right)^6 + \dots \right] \\
&= 2\pi \sqrt{\frac{l}{g}} \left[1 + \underbrace{\mathcal{O}(\phi^2)}_{\text{neglected}} \right] \\
&\stackrel{9}{\approx} 2\pi \sqrt{\frac{l}{g}}
\end{aligned}$$

Which is exactly the time period we had obtained previously.

9. We could have obtained this from Eq. (1.29), by doing a small angle approximation.

$$\begin{aligned}
T = 4t_{1/4} &= 4 \int_0^\phi \frac{d\theta}{\omega \sqrt{2 \left[\left(1 - \frac{\theta^2}{2} \right) - \left(1 - \frac{\phi^2}{2} \right) \right]}} \\
&= 4 \int_0^\phi \frac{d\theta}{\omega \sqrt{\phi^2 - \theta^2}} \\
&= \frac{4}{\omega} \left[\arcsin \left(\frac{\theta}{\phi} \right) \right]_0^\phi \\
&= \frac{4}{\omega} \left[\arcsin(1) - \arcsin(0) \right]
\end{aligned}$$

1.5.3 Magnetic vector potential due to a current carrying loop

Consider a circular wire loop of radius a in the “ xy ” plane carries current I . This is schematically shown in Fig (1.5).

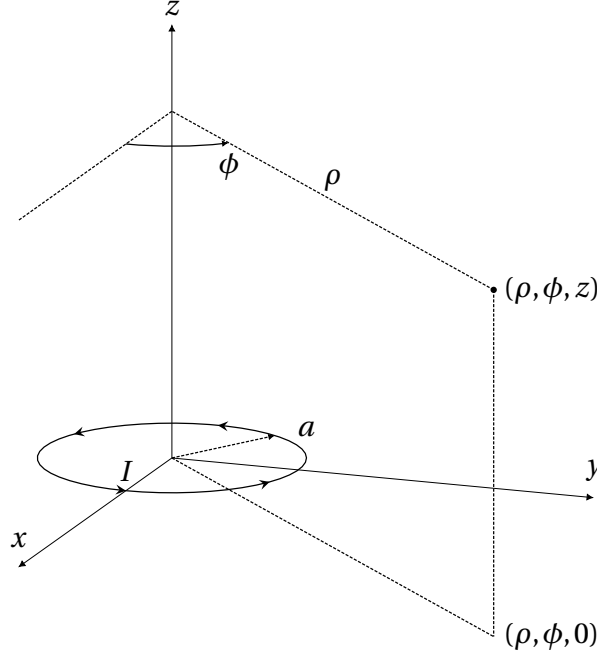


Figure 1.5: Circular wire of radius a in xy plane.

Now the vector potential at any point (ρ, ϕ, z) is given by :

$$A_\phi(\rho, \phi, z) = \frac{\mu_0 I a}{2\pi} \int_0^\pi \frac{\cos \alpha \, d\alpha}{[a^2 + \rho^2 + z^2 - 2a\rho \cos \alpha]^{1/2}} \quad (1.33)$$

We have :

$$\begin{aligned} a^2 + \rho^2 + z^2 - 2a\rho \cos \alpha &= (a + \rho)^2 + z^2 - 2a\rho - 2a\rho \cos \alpha \\ &= (a + \rho)^2 + z^2 - 2a\rho(1 + \cos \alpha) \\ &= (a + \rho)^2 + z^2 - 4a\rho \cos^2 \frac{\alpha}{2} \\ &= [(a + \rho)^2 + z^2] \left[1 - \frac{4a\rho}{(a + \rho)^2 + z^2} \cos^2 \frac{\alpha}{2} \right] \end{aligned}$$

Writing $k^2 = \frac{4a\rho}{(a+\rho)^2+z^2} \Rightarrow (a + \rho)^2 + z^2 = 4a\rho/k^2$, we write Eq. (1.33) as :

$$\begin{aligned} A_\phi(\rho, \phi, z) &= \frac{\mu_0 I a}{2\pi} \int_0^\pi \frac{\cos \alpha \, d\alpha}{\sqrt{\frac{4a\rho}{k^2}} [1 - k^2 \cos^2 \frac{\alpha}{2}]^{1/2}} \\ &= \frac{\mu_0 I k}{4\pi} \sqrt{\frac{a}{\rho}} \int_0^\pi \frac{\cos \alpha \, d\alpha}{\sqrt{1 - k^2 \cos^2 \frac{\alpha}{2}}} \\ &= \frac{\mu_0 I k}{2\pi} \sqrt{\frac{a}{\rho}} \int_0^{\pi/2} \frac{\cos 2\theta \, d\theta}{\sqrt{1 - k^2 \cos^2 \theta}} \end{aligned}$$

$$= \frac{4}{\omega} \cdot \frac{\pi}{2} = \frac{2\pi}{\omega} = 2\pi \sqrt{\frac{l}{g}}$$

$$\begin{aligned}
&= \frac{\mu_0 I k}{2\pi} \sqrt{\frac{a}{\rho}} \int_0^{-\pi/2} \frac{\cos 2\varphi d\varphi}{\sqrt{1-k^2 \sin^2 \varphi}} \quad 2 \\
&= -\frac{\mu_0 I k}{2\pi} \sqrt{\frac{a}{\rho}} \int_0^{\pi/2} \frac{\cos 2\tilde{\phi} d\tilde{\phi}}{\sqrt{1-k^2 \sin^2 \tilde{\phi}}} \quad 3 \\
&= -\frac{\mu_0 I k}{2\pi} \sqrt{\frac{a}{\rho}} \int_0^{\pi/2} \frac{1-2\sin^2 \tilde{\phi} d\tilde{\phi}}{\sqrt{1-k^2 \sin^2 \tilde{\phi}}} \\
&= -\frac{\mu_0 I k}{2\pi} \sqrt{\frac{a}{\rho}} \left[\int_0^{\pi/2} \frac{d\tilde{\phi}}{\sqrt{1-k^2 \sin^2 \tilde{\phi}}} - \frac{2}{k^2} \int_0^{\pi/2} \frac{k^2 \sin^2 \tilde{\phi} - 1 + 1}{\sqrt{1-k^2 \sin^2 \tilde{\phi}}} d\tilde{\phi} \right] \\
&= -\frac{\mu_0 I k}{2\pi} \sqrt{\frac{a}{\rho}} \left[K(k) + \frac{2}{k^2} \int_0^{\pi/2} \sqrt{1-k^2 \sin^2 \tilde{\phi}} d\tilde{\phi} - \frac{2}{k^2} \int_0^{\pi/2} \frac{d\tilde{\phi}}{\sqrt{1-k^2 \sin^2 \tilde{\phi}}} \right] \\
&= -\frac{\mu_0 I k}{2\pi} \sqrt{\frac{a}{\rho}} \left[K(k) + \frac{2}{k^2} E(k) - \frac{2}{k^2} K(k) \right] \\
&= -\frac{\mu_0 I}{\pi k} \sqrt{\frac{a}{\rho}} \times \frac{k^2}{2} \left[\left(1 - \frac{2}{k^2}\right) K(k) + \frac{2}{k^2} E(k) \right] \\
&= -\frac{\mu_0 I}{\pi k} \sqrt{\frac{a}{\rho}} \left[\left(\frac{k^2}{2} - 1\right) K(k) + E(k) \right]
\end{aligned}$$

So finally we have :

$$\boxed{A_\phi(\rho, \phi, z) = \frac{\mu_0 I}{\pi k} \sqrt{\frac{a}{\rho}} \left[\left(1 - \frac{k^2}{2}\right) K(k) - E(k) \right]} \quad (1.34)$$

1. Substitute $\theta = \frac{\alpha}{2}$ which gives $d\alpha = 2d\theta$.
2. Substitute $\theta = \varphi + \frac{\pi}{2}$ which gives $d\varphi = d\theta$.
3. Substitute $\tilde{\phi} = -\varphi$ which gives $d\tilde{\phi} = -d\varphi$.

1.6 Numerical Computation

So far we have obtained a closed form for both $K(k)$ and $E(k)$ as given by Eqs. (1.6) and (1.11) respectively. We have discussed some applications of it, up to some extent. But one question remains, “How one can compute them”. After all this is what we want to do in the first place. All that’s left is to do the “Numerical Computation”. In the following subsections, we will discuss mainly three ways to compute them in detail.

1.6.1 Using the Series

For a given k value, one can naively use the series given by Eqs. (1.6) and (1.11) to compute the values of $K(k)$ and $E(k)$ respectively. Using the recursion relations as given by Eqs. (1.15) and (1.18) helps. Now in practice we can’t add “infinite” number of terms, often we stop after performing the summation up to some finite number of terms(as per our need). Also we have the precision of the computer, by which we are limited. In Table. 1.1 we quoted the values of $K(k)$ for $k = 0.1, \dots, 0.9$ and also no of terms summed. We can easily see that as the k value is approaching 1, we need more and more no of terms to get a fixed precision. This is computationally “inefficient” and “time consuming”.

k	No of terms	$K(k)$
$k = 0.1$	8	1.5747455615173558
$k = 0.2$	11	1.5868678474541662
$k = 0.3$	14	1.6080486199305128
$k = 0.4$	18	1.639999865864511
$k = 0.5$	23	1.6857503548125958
$k = 0.6$	31	1.7507538029157514
$k = 0.7$	43	1.8456939983747227
$k = 0.8$	67	1.9953027776647276
$k = 0.9$	137	2.2805491384227654

Table 1.1: $K(k)$ for various values of k , with $\epsilon = 10^{-15}$ computed using Eq. (1.6).

In Table. 1.2 we have tabulated $E(k)$ values in a similar sense as done before for $K(k)$.

k	No of terms	$E(k)$
$k = 0.1$	8	1.5668619420216683
$k = 0.2$	10	1.5549685462425291
$k = 0.3$	13	1.5348334649232491
$k = 0.4$	16	1.5059416123600406
$k = 0.5$	21	1.4674622093394272
$k = 0.6$	27	1.418083394448724
$k = 0.7$	37	1.3556611355719563
$k = 0.8$	57	1.2763499431699086
$k = 0.9$	112	1.1716970527816197

Table 1.2: $E(k)$ for various values of k , with $\epsilon = 10^{-15}$ computed using Eq. (1.11).

1.6.2 The AGM

Arithmetic–geometric mean also referred as “AGM” is a beautiful concept which states the following :

Consider two real positive numbers x and y , where $x \geq y > 0$. Now do as follows :

$$\begin{aligned} a_0 &= x, & b_0 &= y \\ a_{n+1} &= \frac{a_n + b_n}{2}, & b_{n+1} &= \sqrt{a_n b_n} \end{aligned}$$

Now the fact is that whatever x and y you start with, if you keep doing this again and again, i.e. $a_0, b_0 \rightarrow a_1, b_1 \rightarrow a_2, b_2 \dots$ these two sequences converge to the same number. This limiting number at which both converges to is known as “arithmetic–geometric mean” of x and y , and denoted by $M(x, y)$ or $agm(x, y)$ or $AGM(x, y)$. Mathematically we write :

$$\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} b_n = AGM(x, y)$$

I am giving an example of this, done up to few iterations, after which tolerance of the computer is reached. This is shown in Table. 1.3. I urge you to pause for a moment and start with your own two values for x and y and repeat the same. This is shown in algorithm 1.

Algorithm 1 Arithmetic–Geometric Mean (AGM)

Require: $x > 0$, $y > 0$, tolerance ϵ

```

1:  $a_0 \leftarrow x$ 
2:  $b_0 \leftarrow y$ 
3:  $n \leftarrow 0$ 
4: while  $|a_n - b_n| > \epsilon$  do
5:    $a_{n+1} \leftarrow \frac{a_n + b_n}{2}$ 
6:    $b_{n+1} \leftarrow \sqrt{a_n b_n}$ 
7:    $n \leftarrow n + 1$ 
8: end while
9: return  $AGM(x, y) \approx a_n$ 

```

n	a_n	b_n	$ a_n - b_n $
$n = 0$	1	0.5	0.5
$n = 1$	0.75	0.7071067811865476	0.04289321881345243
$n = 2$	0.7285533905932737	0.7282376575609851	0.00031573303228860095
$n = 3$	0.7283955240771294	0.7283955069697775	$1.710735186843948e - 08$
$n = 4$	0.7283955155234534	0.7283955155234534	0.0

Table 1.3: AGM for $x = 1$ and $y = \frac{1}{2}$ with $\epsilon = 10^{-15}$.

Connecting it with $K(k)$, which was discovered by “Carl Friedrich Gauss”, very very long ago, we got :

$$K(k) = \frac{\pi}{2AGM(1, \sqrt{1-k^2})} \quad (1.35)$$

So the task of computing $K(k)$ simplifies to find the “AGM” of two numbers “1” and “ $\sqrt{1-k^2}$ ”, which is very easy to find. This procedure converges very fast and practically best for computing $K(k)$. In algorithm 2 this procedure is shown. We also give a tabular representation example by Table. 1.4.

10. See Sec. 1.7.2

Algorithm 2 Computation of the Complete Elliptic Integral $K(k)$ using AGM**Require:** $0 \leq k < 1$, tolerance ϵ

```

1:  $a_0 \leftarrow 1$ 
2:  $b_0 \leftarrow \sqrt{1 - k^2}$ 
3:  $n \leftarrow 0$ 
4: while  $|a_n - b_n| > \epsilon$  do
5:    $a_{n+1} \leftarrow \frac{a_n + b_n}{2}$ 
6:    $b_{n+1} \leftarrow \sqrt{a_n b_n}$ 
7:    $n \leftarrow n + 1$ 
8: end while
9:  $M \leftarrow a_n$ 
10: return  $K(k) = \frac{\pi}{2M}$ 

```

n	a_n	b_n	$ a_n - b_n $
$n = 0$	1	0.8660254037844386	0.1339745962155614
$n = 1$	0.9330127018922193	0.9306048591020996	0.002407842790119741
$n = 2$	0.9318087804971594	0.9318080027478182	$7.777493412630676e - 07$
$n = 3$	0.9318083916224889	0.9318083916224076	$8.126832540256146e - 14$
$n = 4$	0.9318083916224482	0.9318083916224482	0.0

Table 1.4: AGM for $x = 1$ and $y = \sqrt{1 - k^2}$, with $k = 0.5$. Which gives $K(k = 0.5) = 1.685750354812596$ with $\epsilon = 10^{-15}$.

In Table 1.5 we quoted the values of $K(k)$ for $k = 0.1, \dots, 0.9$, with their corresponding AGM values and no of iterations performed. We see that no of iterations are not increasing much as we approach close to 1. This is much suitable for computing $K(k)$.

k	AGM	Iterations	$K(k)$
$k = 0.1$	0.9974921442428744	3	1.574745561517356
$k = 0.2$	0.9898721744945217	3	1.5868678474541664
$k = 0.3$	0.9768338514931061	3	1.6080486199305126
$k = 0.4$	0.9578027166281903	4	1.6399998658645112
$k = 0.5$	0.9318083916224482	4	1.685750354812596
$k = 0.6$	0.8972114321150411	4	1.7507538029157523
$k = 0.7$	0.8510599959571328	4	1.8456939983747236
$k = 0.8$	0.7872471007299109	4	1.9953027776647299
$k = 0.9$	0.6887798646080748	4	2.280549138422769

Table 1.5: $K(k)$ for various values of k , with $\epsilon = 10^{-15}$ computed using AGM.

So far we have talked about how to calculate $K(k)$ using “AGM”. Now we will talk about how we can compute $E(k)$. I will discuss only the outline. For a given k do the following :

$$\begin{aligned} a_0 &= 1, & b_0 &= \sqrt{1-k^2}, & c_0 &= k \\ a_{n+1} &= \frac{a_n + b_n}{2}, & b_{n+1} &= \sqrt{a_n b_n}, & c_{n+1} &= \frac{a_n - b_n}{2} \end{aligned}$$

The value of $E(k)$ is given by :

$$E(k) = K(k) \left[1 - \sum_{r=0}^{\infty} 2^{r-1} c_r^2 \right] \quad (1.36)$$

Where $K(k)$ is given by Eq. (1.35). In each iteration a_n and b_n gets closer and closer and $c_n \rightarrow 0$ as the precision is reached. In practice we only have to add only finite number of c_n terms in Eq. (1.36). We have shown in algorithm 3 and also a tabular representation in Tables. 1.6 and 1.7.

Algorithm 3 Computation of the Complete Elliptic Integral $E(k)$ using AGM

Require: $0 \leq k < 1$, tolerance ϵ

```

1:  $a_0 \leftarrow 1$ 
2:  $b_0 \leftarrow \sqrt{1-k^2}$ 
3:  $c_0 \leftarrow k$ 
4:  $S_0 \leftarrow 0$ 
5:  $n \leftarrow 0$ 
6: while  $|c_n| > \epsilon$  do
7:    $a_{n+1} \leftarrow \frac{a_n + b_n}{2}$ 
8:    $b_{n+1} \leftarrow \sqrt{a_n b_n}$ 
9:    $c_{n+1} \leftarrow \frac{a_n - b_n}{2}$ 
10:   $S_{n+1} \leftarrow S_n + 2^{n-1} c_n^2$ 
11:   $n \leftarrow n + 1$ 
12: end while
13:  $M \leftarrow a_n$ 
14:  $S \leftarrow S_n$ 
15:  $K(k) \leftarrow \frac{\pi}{2M}$ 
16:  $E(k) \leftarrow K(k)[1 - S]$ 
17: return  $(K(k), E(k))$ 

```

n	a_n	b_n	c_n	S_n
$n = 0$	1	0.714142842854285	0.7	0.0
$n = 1$	0.8570714214271424	0.845069726622771	0.1429285785728575	0.24499999999999997
$n = 2$	0.8510705740249567	0.8510494179550474	0.006000847402185705	0.26542857857285745
$n = 3$	0.851059995990002	0.8510599959242635	$1.0578034954633608e-05$	0.2655005989119461
$n = 4$	0.8510599959571328	0.8510599959571328	$3.2869262867052385e-11$	0.2655005993595254
$n = 5$	0.8510599959571328	0.8510599959571328	0.0	0.2655005993595254

Table 1.6: For $k = 0.7$ we obtain $M = 0.8510599959571328$ and $S = 0.2655005993595254$ which gives $K(k = 0.7) = 1.8456939983747236$ and $E(k = 0.7) = 1.3556611355719554$ with $\epsilon = 10^{-15}$.

k	AGM	Iterations	S	$K(k)$	$E(k)$
$k = 0.1$	0.9974921442428744	4	0.005006281451646927	1.574745561517356	1.5668619420216683
$k = 0.2$	0.9898721744945217	4	0.020102052771951644	1.5868678474541664	1.5549685462425296
$k = 0.3$	0.9768338514931061	4	0.045530436144665544	1.6080486199305126	1.534833464923249
$k = 0.4$	0.9578027166281903	4	0.08174284418846783	1.6399998658645112	1.5059416123600404
$k = 0.5$	0.9318083916224482	5	0.12949019696183656	1.685750354812596	1.4674622093394272
$k = 0.6$	0.8972114321150411	5	0.19001552811879663	1.7507538029157523	1.418083394448724
$k = 0.7$	0.8510599959571328	5	0.2655005993595254	1.8456939983747236	1.3556611355719554
$k = 0.8$	0.7872471007299109	5	0.36032267510611804	1.9953027776647299	1.2763499431699066
$k = 0.9$	0.6887798646080745	5	0.4862215275080805	2.2805491384227703	1.171697052781614

Table 1.7: $K(k)$ and $E(k)$ for various values of k , with $\epsilon = 10^{-15}$ computed using Eqs. (1.35) and (1.36).

1.6.3 Alternative way to get $E(k)$

For a given k value we numerically calculate the derivative of $K(k)$ i.e.

$$\frac{dK}{dk} \approx \frac{K(k+\delta) - K(k-\delta)}{2\delta} + \mathcal{O}(\delta^2) \quad (1.37)$$

Using this derivative value and Eq. (1.47) we get the value of $E(k)$ as follows :

$$E(k) = (1 - k^2) \left[k \frac{K(k+\delta) - K(k-\delta)}{2\delta} + K(k) \right] \quad (1.38)$$

We can take the value of $\delta = \sqrt{\epsilon}$. In Table. 1.8 we have compared the values of $E(k)$ obtained this way and by the way described previously.

k	$E(k)$	$E(k)_{dv}$	$ E(k) - E(k)_{dv} $
$k = 0.1$	1.5668619420216683	1.566861942359351	$3.376827706347285e - 10$
$k = 0.2$	1.5549685462425296	1.5549685465188992	$2.763695938767796e - 10$
$k = 0.3$	1.534833464923249	1.5348334641734278	$7.498210941037087e - 10$
$k = 0.4$	1.5059416123600404	1.5059416135594543	$1.1994139015314431e - 09$
$k = 0.5$	1.4674622093394272	1.467462210731052	$1.3916248153833521e - 09$
$k = 0.6$	1.418083394448724	1.4180833956089043	$1.1601801741534246e - 09$
$k = 0.7$	1.3556611355719554	1.3556611367061988	$1.1342433658967366e - 09$
$k = 0.8$	1.2763499431699066	1.2763499431782432	$8.3366646919103e - 12$
$k = 0.9$	1.171697052781614	1.1716970521371235	$6.444904609992363e - 10$

Table 1.8: $E(k)$ for various values of k , with $\epsilon = 10^{-15}$ computed using Eqs. (1.36) and (1.38). Here we have computed $K(k)$ using Eq. (1.35). The subscript “ dv ” signifies that it has been computed by the derivative formula given by Eq. (1.38).

1.6.4 Performing the Integration Numerically

By all means what we can do is to perform the integration in Eqs. (1.4) and (1.10). There are several ways to compute the integral, like “Trapezoidal Rule”, “Simpson’s Rule” etc. Here we will discuss another method for computing the integration which is “Gauss–Legendre Quadrature” which is discussed below.

Consider a function $f(x)$ defined in the interval $[-1, 1]$ and we approximate the integral like the following :

$$\int_{-1}^1 f(x) dx \approx \sum_{i=1}^n w_i f(x_i) \quad (1.39)$$

Where :

1. n is the number of sample points used.
2. w_i are quadrature weights, where they are given by $w_i = \frac{2}{(1-x_i^2)[P'_n(x_i)]^2}$.¹¹
3. x_i are the roots of the n th Legendre polynomial, $\implies P_n(x_i) = 0$.

We use Eqs. (1.4a) and (1.10a) to take the functional form but first we have to change the interval to $[-1, 1]$.¹² Performing the substitution $\theta = \frac{\pi}{4}(x+1)$ in Eqs. (1.4a) and (1.10a) we get :

$$K(k) = \frac{\pi}{4} \int_{-1}^1 \left[1 - k^2 \sin^2 \left[\frac{\pi}{4}(x+1) \right] \right]^{-\frac{1}{2}} dx$$

$$E(k) = \frac{\pi}{4} \int_{-1}^1 \left[1 - k^2 \sin^2 \left[\frac{\pi}{4}(x+1) \right] \right]^{\frac{1}{2}} dx$$

In Table. 1.9 we have tabulated the values of $K(k)$ and $E(k)$ for a fixed value of $k = 0.3$ with no of sample points n as a variable.

n	$K(k)$	$E(k)$
$n = 2$	1.608209656003143	1.5347822055615428
$n = 4$	1.6080493015991812	1.5348332419413315
$n = 8$	1.608048619929695	1.5348334649233568
$n = 16$	1.6080486199305124	1.534833464923249
$n = 32$	1.6080486199305128	1.534833464923249
$n = 64$	1.6080486199305128	1.534833464923249

Table 1.9: $K(k), E(k)$ for $k = 0.3$, with no of sample points n as a variable.

¹¹. The superscript $'$ means derivative w.r.t. x and $P'_n(x_i) = \left. \frac{dP_n(x)}{dx} \right|_{x=x_i}$

¹². If the limits are from $x = a$ to $x = b$, we do the following substitution :

$$x = \frac{b-a}{2} y + \frac{b+a}{2} \implies y = -1 + 2 \frac{x-b}{b-a} \quad (1.40)$$

1.7 Additional Calculations

1.7.1 Evaluation of Integral

Using the identity of “beta functions” we can write :

$$\begin{aligned}\beta\left(\frac{p+1}{2}, \frac{q+1}{2}\right) &= \frac{1}{2} \int_0^{\frac{\pi}{2}} \sin^p \theta \cos^q \theta d\theta \\ &= \frac{\Gamma\left(\frac{p+1}{2}\right) \times \Gamma\left(\frac{q+1}{2}\right)}{2\Gamma\left(\frac{p+q+2}{2}\right)}\end{aligned}$$

With $2r = p$ and $q = 0$ the last equation becomes:

$$\begin{aligned}\int_0^{\frac{\pi}{2}} \sin^{2r} \theta d\theta &= \frac{\Gamma\left(\frac{2r+1}{2}\right) \times \Gamma\left(\frac{1}{2}\right)}{2\Gamma\left(\frac{2r+2}{2}\right)} \\ &= \frac{\Gamma\left(\frac{2r+1}{2}\right) \times \Gamma\left(\frac{1}{2}\right)}{2\Gamma(r+1)}\end{aligned}\tag{1.41}$$

Now we have :

$$\begin{aligned}\Gamma\left(\frac{2r+1}{2}\right) &= \Gamma\left(r + \frac{1}{2}\right) \\ &= \Gamma\left(r - \frac{1}{2} + 1\right) \\ &= \left(r - \frac{1}{2}\right) \cdot \Gamma\left(r - \frac{1}{2}\right) \\ &= \left(r - \frac{1}{2}\right) \cdot \left(r - \frac{3}{2}\right) \cdot \Gamma\left(r - \frac{3}{2}\right) \\ &= \left(r - \frac{1}{2}\right) \cdot \left(r - \frac{3}{2}\right) \cdot \left(r - \frac{5}{2}\right) \cdot \Gamma\left(r - \frac{5}{2}\right) \\ &\quad \vdots \\ &\quad \vdots \\ &= \underbrace{\left(r - \frac{1}{2}\right) \left(r - \frac{3}{2}\right) \left(r - \frac{5}{2}\right) \cdots \frac{3}{2} \cdot \frac{1}{2}}_{r \text{ terms}} \Gamma\left(\frac{1}{2}\right) \\ &= \underbrace{\left(\frac{2r-1}{2}\right) \left(\frac{2r-3}{2}\right) \left(\frac{2r-5}{2}\right) \cdots \frac{3}{2} \cdot \frac{1}{2}}_{r \text{ terms}} \Gamma\left(\frac{1}{2}\right) \\ &= \frac{(2r-1)(2r-3)(2r-5) \cdots 3 \cdot 1}{\underbrace{2 \cdot 2 \cdot 2 \cdots 2}_{r \text{ times}}} \Gamma\left(\frac{1}{2}\right) \\ &= \frac{(2r-1)!!}{2^r} \Gamma\left(\frac{1}{2}\right)\end{aligned}\tag{1.42}$$

$$\begin{aligned}&= \frac{(2r)(2r-1)(2r-2)(2r-3) \cdots 3 \cdot 2 \cdot 1}{2^r \times \underbrace{(2r)(2r-2) \cdots 2}_{r \text{ terms}}} \Gamma\left(\frac{1}{2}\right) \\ &= \frac{(2r)!}{2^r \times (2r)!!} \Gamma\left(\frac{1}{2}\right) = \frac{(2r)!}{2^{2r} r!} \Gamma\left(\frac{1}{2}\right)\end{aligned}\tag{1.43}$$

Using Eqs. (1.41) and (1.43) we get :

$$\int_0^{\frac{\pi}{2}} \sin^{2r} \theta d\theta = \frac{\pi}{2} \cdot \frac{(2r)!}{2^{2r} (r!)^2}$$

Also using Eqs. (1.41) and (1.42) we write this as :

$$\int_0^{\frac{\pi}{2}} \sin^{2r} \theta \, d\theta = \frac{\pi}{2} \cdot \frac{(2r-1)!!}{2^r r!} = \frac{\pi}{2} \cdot \frac{(2r-1)!!}{(2r)!!}$$

1.7.2 Connecting AGM to Elliptic Integral

Consider two positive real numbers x and y such that $x \geq y > 0$. Then we must have :

$$I(x, y) = \int_0^{\pi/2} \frac{d\theta}{\sqrt{x^2 \cos^2 \theta + y^2 \sin^2 \theta}} = I\left(\frac{x+y}{2}, \sqrt{xy}\right) \quad (1.44)$$

See [the proof here](#). Now consider that $x = a_0$ and $y = b_0$, then $I(a_0, b_0) = I(a_1, b_1)$. Where the symbols carry the same meaning as in Sec. 1.6.2. Now we can continue doing this, meaning $I(a_1, b_1) = I(a_2, b_2) = \dots = I(a_n, b_n)$. Afterwards we will reach $a_\infty = b_\infty = AGM(x, y)$. Where both the terms are equal. Now if both terms are equal then the integral is simply $I(a_\infty, b_\infty) = \frac{1}{a_\infty} \int_0^{\pi/2} d\theta = \frac{\pi}{2a_\infty}$. So we finally have :

$$\int_0^{\pi/2} \frac{d\theta}{\sqrt{x^2 \cos^2 \theta + y^2 \sin^2 \theta}} = \frac{\pi}{2AGM(x, y)} \quad (1.45)$$

Now if we put $x = 1$ and $y = \sqrt{1-k^2}$, then the denominator becomes $\sqrt{\cos^2 \theta + (1-k^2) \sin^2 \theta} = \sqrt{1-k^2 \sin^2 \theta}$. Which is basically the form of “Complete Elliptic Integral of first kind.” So by this we got Eq. (1.35).

1.7.3 An Identity

The identity is given by :

$$\begin{aligned} \frac{d}{dk} K(k) &= \frac{E(k)}{k(1-k^2)} - \frac{K(k)}{k} \\ \Rightarrow E(k) &= (1-k^2) \left[k \frac{dK(k)}{dk} + K(k) \right] \end{aligned}$$

We will prove it this way. Using Eq. (1.4a) we have :

$$\begin{aligned} \frac{d}{dk} K(k) &= \frac{d}{dk} \left[\int_0^{\pi/2} \frac{d\theta}{(1-k^2 \sin^2 \theta)^{\frac{1}{2}}} \right] \\ &= \int_0^{\pi/2} \frac{k \sin^2 \theta}{(1-k^2 \sin^2 \theta)^{\frac{3}{2}}} d\theta \\ &= \frac{1}{k} \int_0^{\pi/2} \frac{k^2 \sin^2 \theta}{(1-k^2 \sin^2 \theta)^{\frac{3}{2}}} d\theta \\ &= \frac{1}{k} \int_0^{\pi/2} \frac{1 - (1-k^2 \sin^2 \theta)}{(1-k^2 \sin^2 \theta)^{\frac{3}{2}}} d\theta \\ &= \frac{1}{k} \int_0^{\pi/2} \frac{d\theta}{(1-k^2 \sin^2 \theta)^{\frac{3}{2}}} - \frac{1}{k} \int_0^{\pi/2} \frac{d\theta}{(1-k^2 \sin^2 \theta)^{\frac{1}{2}}} \\ &= \left[\frac{1}{k} \int_0^{\pi/2} (1-k^2 \sin^2 \theta)^{-\frac{3}{2}} d\theta \right] - \frac{K(k)}{k} \end{aligned}$$

We will now evaluate the following :

$$\begin{aligned}
I &= (1 - k^2) \int_0^{\pi/2} (1 - k^2 \sin^2 \theta)^{-\frac{3}{2}} d\theta \\
&= (1 - k^2) \int_0^{\pi/2} d\theta \left[1 + \sum_{r=1}^{\infty} \frac{(2r+1)!!}{(2r)!!} (k^2 \sin^2 \theta)^r \right] \\
&= (1 - k^2) \left[\int_0^{\pi/2} d\theta + \left(\sum_{r=1}^{\infty} \frac{(2r-1)!!}{(2r)!!} k^{2r} \int_0^{\pi/2} \sin^{2r} \theta d\theta \right) \right] \\
&= (1 - k^2) \left[\frac{\pi}{2} + \frac{\pi}{2} \left(\sum_{r=1}^{\infty} \frac{(2r+1)!!}{(2r)!!} \times \frac{(2r-1)!!}{(2r)!!} k^{2r} \right) \right] \\
&= (1 - k^2) \cdot \frac{\pi}{2} \left[1 + \sum_{r=1}^{\infty} \frac{(2r+1)!!}{(2r)!!} \times \frac{(2r-1)!!}{(2r)!!} k^{2r} \right] \\
&= \frac{\pi}{2} \left[1 - k^2 + \sum_{r=1}^{\infty} \frac{(2r+1)!!}{(2r)!!} \times \frac{(2r-1)!!}{(2r)!!} k^{2r} - \sum_{r=1}^{\infty} \frac{(2r+1)!!}{(2r)!!} \times \frac{(2r-1)!!}{(2r)!!} k^{2(r+1)} \right] \\
&= \frac{\pi}{2} \left[1 - k^2 + \frac{3!!}{2!!} \times \frac{1!!}{2!!} k^2 + \sum_{r=2}^{\infty} \frac{(2r+1)!!}{(2r)!!} \times \frac{(2r-1)!!}{(2r)!!} k^{2r} - \sum_{r=2}^{\infty} \frac{(2r-1)!!}{(2r-2)!!} \times \frac{(2r-3)!!}{(2r-2)!!} k^{2r} \right] \\
&= \frac{\pi}{2} \left[1 - k^2 + \frac{3}{4} k^2 + \sum_{r=2}^{\infty} \left(\frac{(2r-1)!!}{(2r)!!} \right)^2 \times \left[(2r+1) - \frac{4r^2}{(2r-1)} \right] k^{2r} \right] \\
&= \frac{\pi}{2} \left[1 - \frac{1}{4} k^2 + \sum_{r=2}^{\infty} \left(\frac{(2r-1)!!}{(2r)!!} \right)^2 \times \left(\frac{4r^2 - 1 - 4r^2}{(2r-1)} \right) k^{2r} \right] \\
&= \frac{\pi}{2} \left[1 - \frac{1}{4} k^2 - \sum_{r=2}^{\infty} \left(\frac{(2r-1)!!}{(2r)!!} \right)^2 \times \frac{k^{2r}}{(2r-1)} \right] \\
&= E(k)
\end{aligned}$$

Which gives :

$$\boxed{\int_0^{\pi/2} (1 - k^2 \sin^2 \theta)^{-\frac{3}{2}} d\theta = \frac{E(k)}{(1 - k^2)}} \quad (1.46)$$

So finally we have the identity :

$$\boxed{\frac{d}{dk} K(k) = \frac{1}{k} \left[\frac{E(k)}{(1 - k^2)} - K(k) \right] \Rightarrow E(k) = (1 - k^2) \left[k \frac{dK(k)}{dk} + K(k) \right]} \quad (1.47)$$

1.8 Type Setting Notes

- (a) This document has been produced using $\text{\LaTeX}$ typesetting.
- (b) The font used is *fourier*. To use this font, include the line `\usepackage{fourier}` in the preamble.
- (c) All the figures have been generated using the TikZ package.
- (d) Fig (1.1) is generated by Pgfplots package.
- (e) The corresponding .tex files to that are used to generate the figures are [available at this link](#).
- (f) Also two .txt files are available at the same link that were used to generate Fig (1.1).
- (g) A jupyter notebook is [available at this link](#) that had been used to generate all the tables , along with the necessary codes written in python .